

QDTraj

Exploration of Diverse Trajectory Primitives for Articulated Objects Robotic Manipulation

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Abstract: QDTraj is a method that generates diverse low-level trajectory primitives for manipulating articulated objects by leveraging Quality-Diversity algorithms.

Project Webpage: https://kappel.web.isir.upmc.fr/trajectory_primitive_website/, **Contact :** kappel@isir.upmc.fr

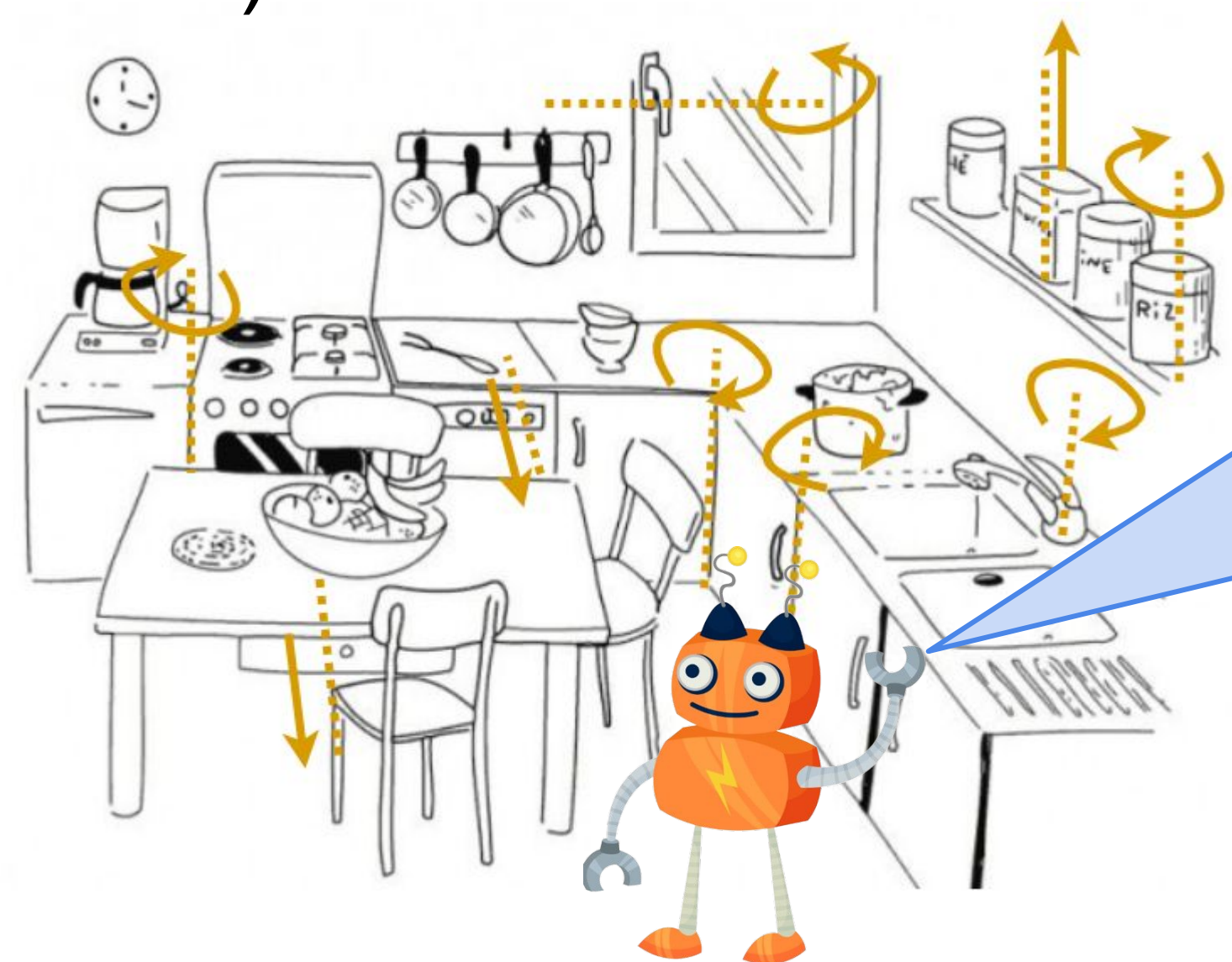
Introduction

Challenges in prehensile robotic manipulation

- Sparse reward, open-ended learning, generalization capabilities, physical interactions.

Articulated objects Manipulation

- Articulated object : An entity composed of multiple rigid parts connected by joints (prismatic and revolute).



How can I learn to manipulate articulated objects ?

- PartNetMobility: Dataset of 2,346 articulated objects [1].

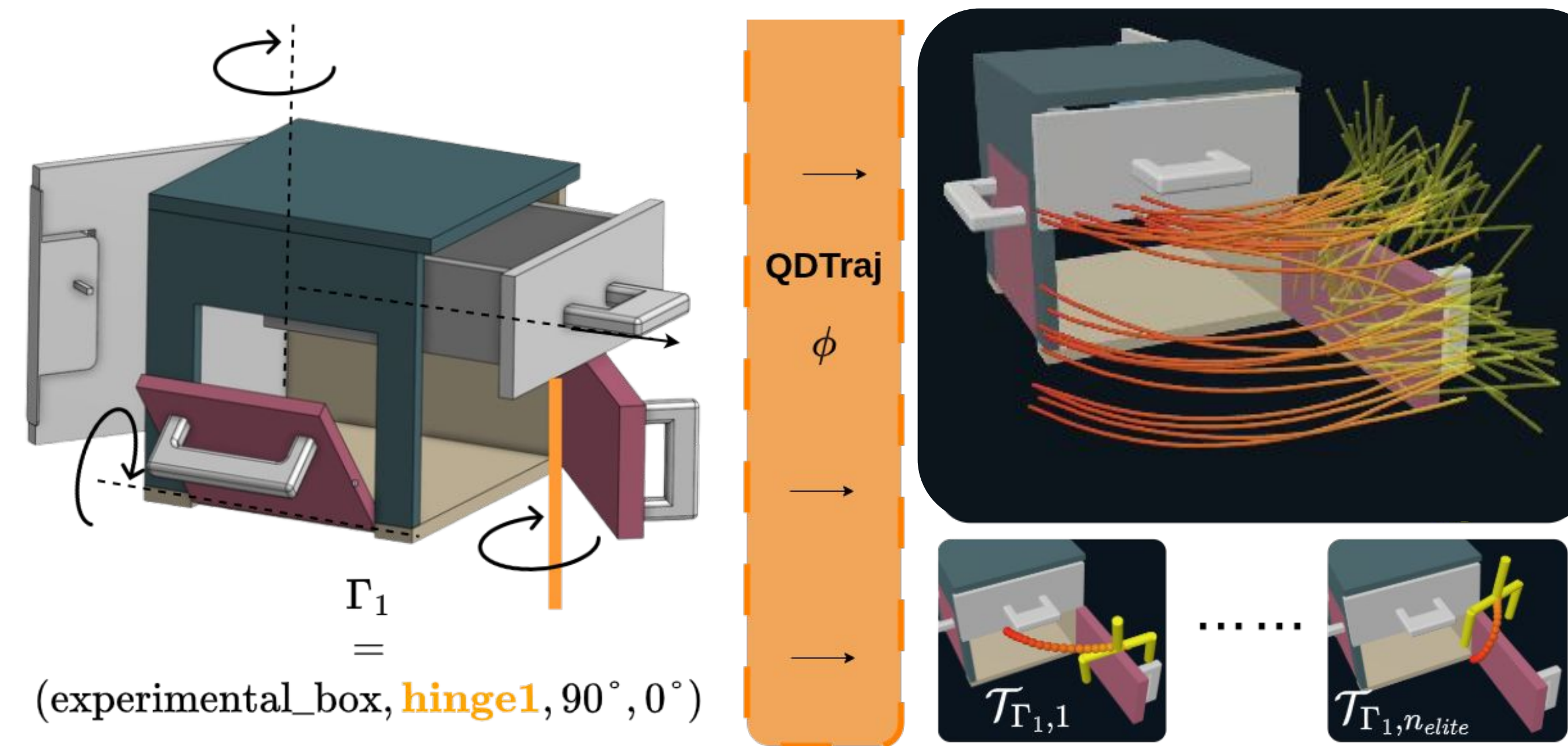
Methods

Movement primitive representation

- *Action Primitive (Input)* = A robot manipulation task for activating a single object articulation.

$$\Gamma = (\text{object}, \text{articulation} + \text{index}, s_{\text{init}}, s_{\text{target}}) \in \mathcal{X} \times \mathcal{A} \times \mathbb{R} \times \mathbb{R}$$

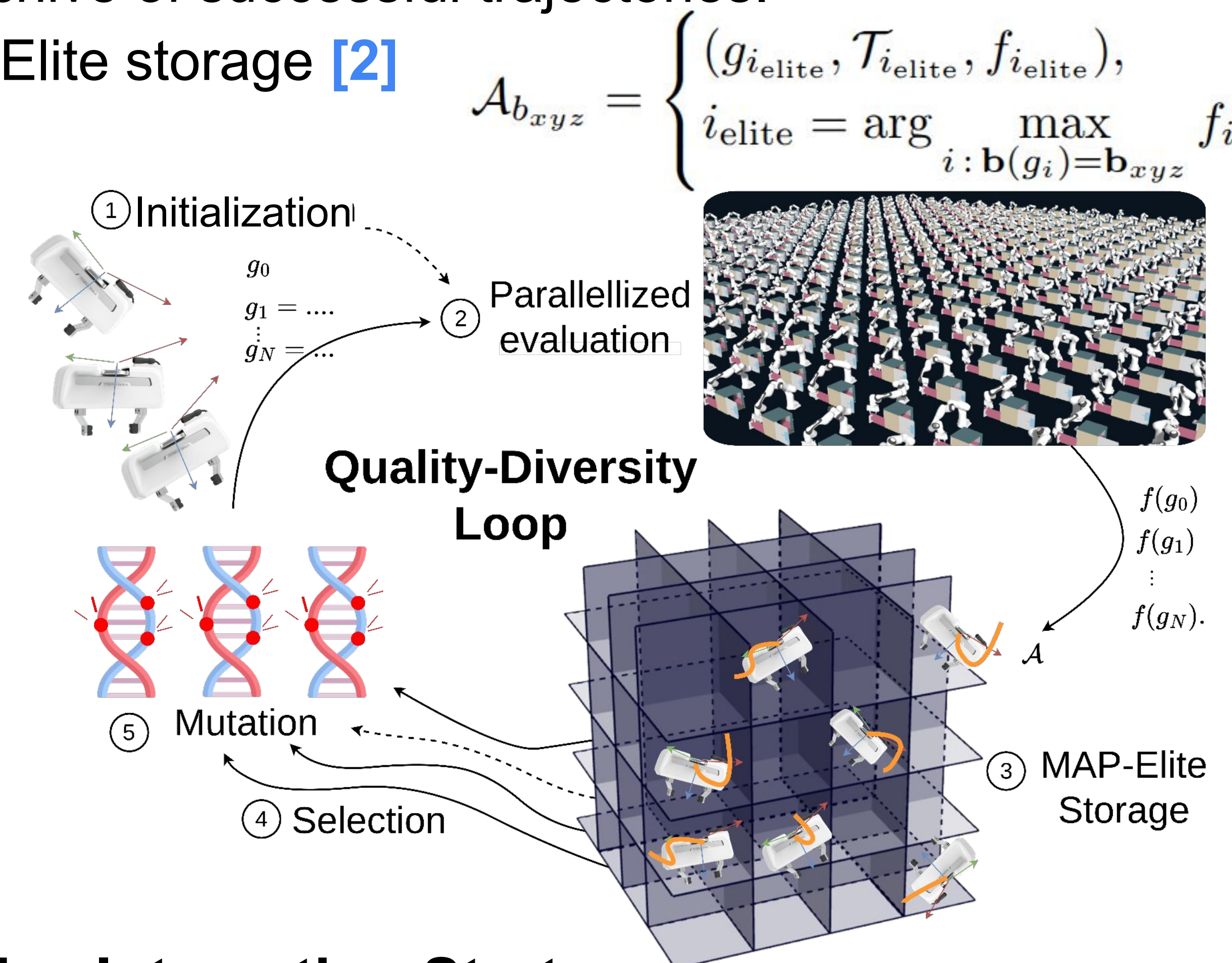
- *Set of Trajectory Primitives (output)* = Sequence of expert trajectories to manipulate a given object articulation.



$$\mathcal{T}_{\Gamma} = \left\{ \mathcal{F}_{ee/object}^{(k)} \right\}_{k=0}^N$$

Learning Algorithm

- QDTraj is an evolutionary algorithm that generates an archive of successful trajectories.
- MAP-Elite storage [2]

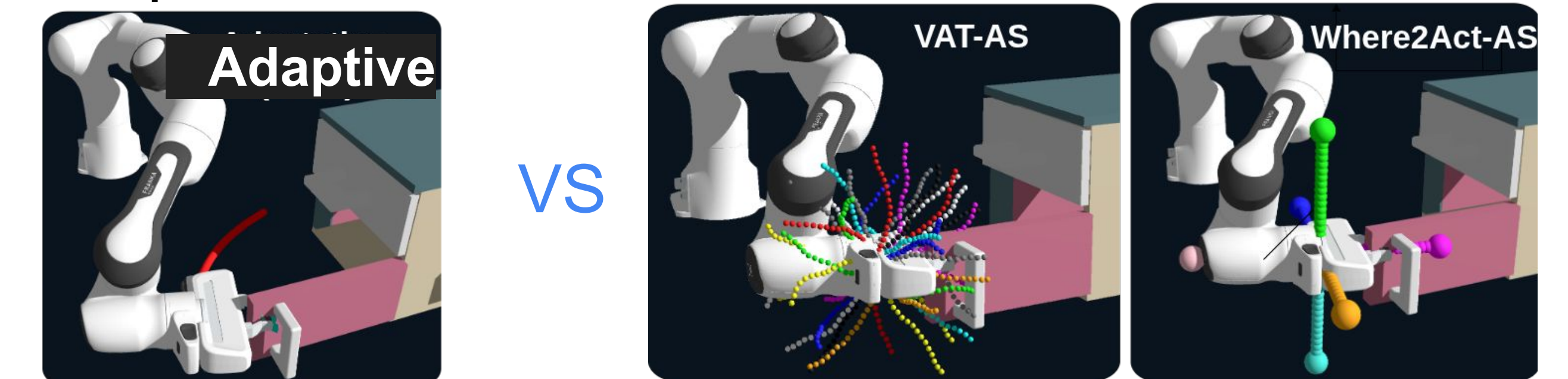


Adaptive Interaction Strategy

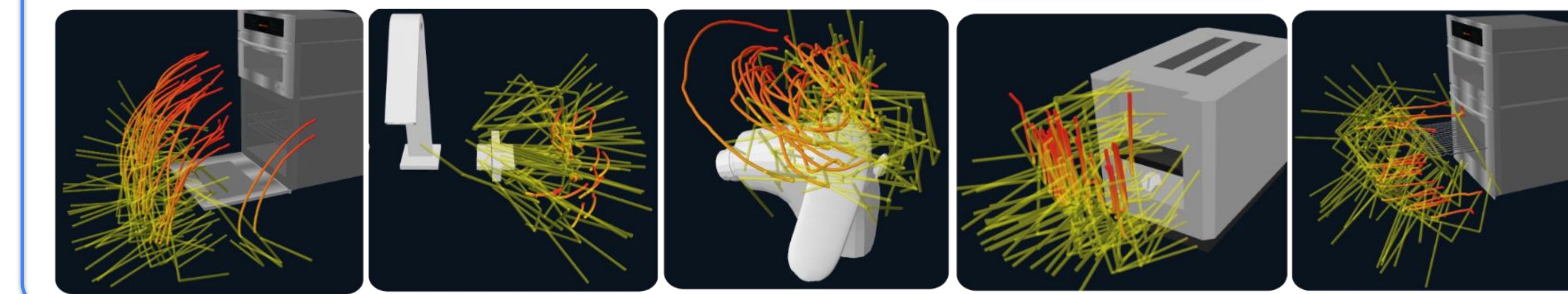
- Adaptive mode = the object guides the robot motion.
- Robot joint gains → Low, Object joint gains → High

Experiments

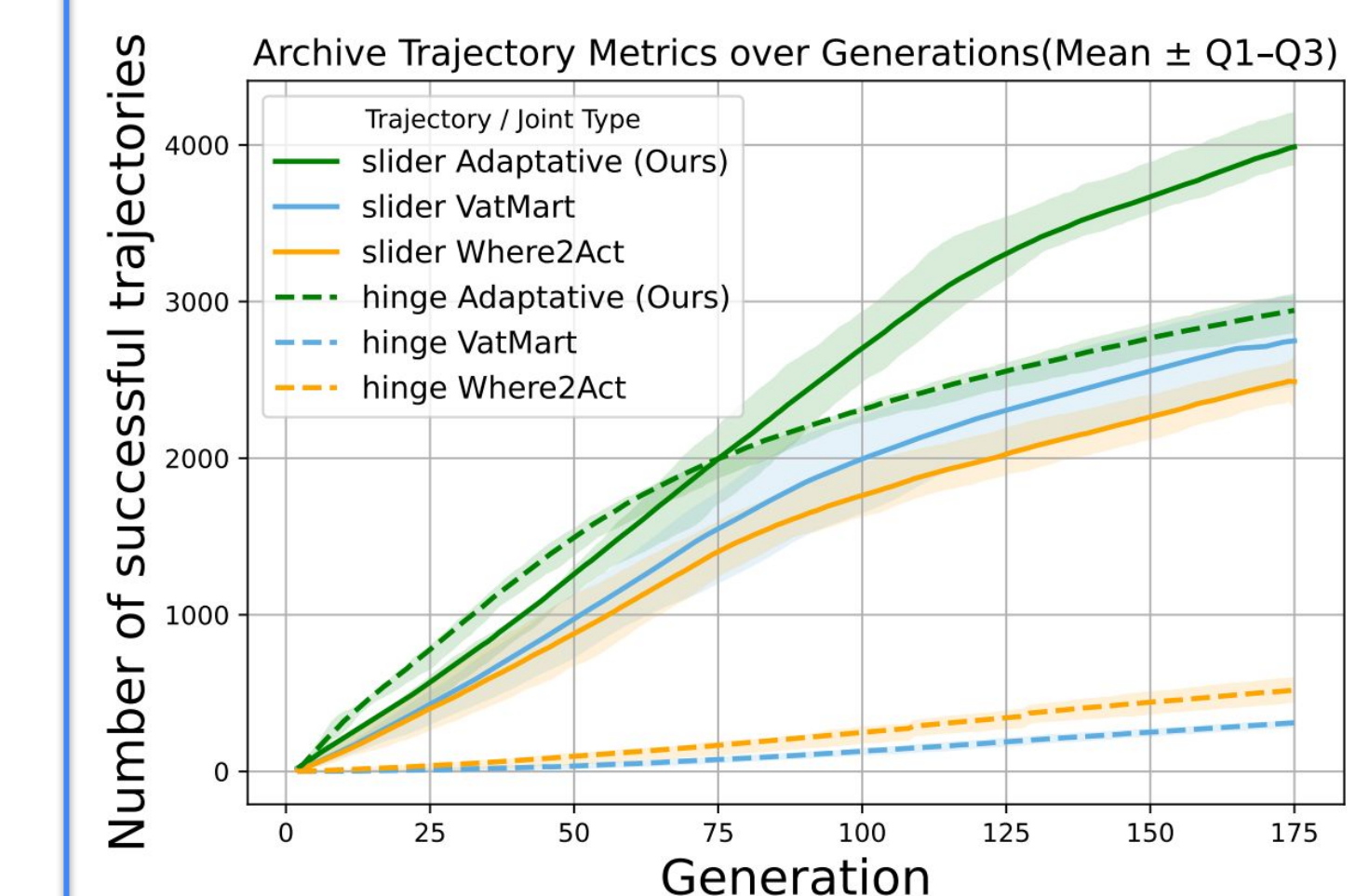
- Grasp strategy ablation study & interaction strategy comparison.



- 3 experimental action primitives + 30 action primitives across 10 PartNet-Mobility objects



Results



- Sampling strategy: ~5X more diverse trajectories
- Interaction policy: ~2.3X more diverse

- Spatial diversity in the trajectories found by QDTraj.
- Non-intuitive trajectories & Non-reversible trajectories

